

HASSAN MAHMOOD

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EDUCATION

Northeastern University

Ph.D. Computer Science (Master's Degree Awarded in 2024)

Advisor: Dr. Ehsan Elhamifar

Research Interests: Generative Models, Efficient Adversarial Attacks and Robustness, Computer Vision

Boston, MA

Jan. 2027 (Expected)

National University of Sciences and Technology

Bachelor of Software Engineering

Advisor: Dr. Faisal Shafait

Research Interests: Computer Vision, Real-time Vehicle Detection and Localization

Islamabad, Pakistan

July 2018

PUBLICATIONS

- **Hassan Mahmood**, Alina Oprea, Ehsan Elhamifar, "Clean-Label Backdoor Attacks for Diffusion Models", (**under review, NeurIPS 2026**)
- **Hassan Mahmood**, Ehsan Elhamifar, "Compositional Targeted Multi-Label Universal Perturbations", **CVPR 2025**. [PDF] [Code]
Workshop Spotlight presentation — *Uncertainty Quantification for Computer Vision Workshop* (CVPR 2025).
- **Hassan Mahmood**, Ehsan Elhamifar, "Semantic-Aware Multi-Label Adversarial Attacks", **CVPR 2024**. [PDF] [Code]
- S. Qasim, **Hassan Mahmood**, Faisal Shafait, "Rethinking Table Recognition using Graph Neural Networks", **ICDAR 2019**. [PDF] [Code]

WORK EXPERIENCE

Ph.D. Intern

Google Cloud AI

May 2026 – Aug. 2026

Seattle, WA

Project: Multi-Agent LLM System for Fleet-Scale Root-Cause Analysis

- Developed and shipped specialized agents in a **hierarchical multi-agent** system performing automated root-cause analysis on security-update failures across VM fleets – each subagent queries its own log sources and separates the originating fault from downstream symptoms via temporal event ordering, while an orchestrator reconciles into a single verdict for **SRE** remediation.
- Scaled the system from single-machine investigation to continuous fleet-wide coverage across **~14M VMs**, cutting mean investigation time **16h → ~2h (8×)** per **10K machines** and inconclusive root-cause rate **20% → ~8%** against the prior production system; deployed to **SRE on-call**.

Graduate Research Assistant

NOVA Lab, Northeastern University

Aug. 2020 – Present

Boston, MA

Project: Action-State Inference in Multi-Task Egocentric Videos with Vision-Language Models (ongoing work)

- **Problem**: Inferring which actions of each task are complete at *any* point in **multi-task egocentric videos** (MTTAS dataset, up to 3 interleaved tasks) requires the full action history, which far exceeds a VLM's context limit. Uniform sampling allocates by duration – oversampling long actions and leaving short ones with **zero frames**, so they are never recognized as completed.
- **Solution**: Built an action-coverage-aware pipeline that jointly selects frames and adaptively merges tokens under a **strict 3K visual token budget**, using no preset segment rules since the action count is unknown in advance. Recovers completion state for past actions of any queried task, raising **F1 from 0.68 to 0.73** over uniform sampling at equal budget.

Project: Clean-Label Backdoor Attacks on Diffusion Models (NeurIPS 2026, under review)

- **Problem**: Under a **data-only threat model**, naive small-perturbation poisons are low-density samples that the denoising process treats as noise and effectively ignores. Making them effective requires optimizing the poisons *through* the diffusion training dynamics, which is intractable to do exactly (**exhausts an H100's 94 GB** for even a single sample).
- **Solution**: Built **DDPF**, the first differentiable bilevel formulation for data-only diffusion poisoning, architecture-agnostic across attacker objectives, in which poisons are optimized so that standard diffusion training learns **corrupted score estimates** on them. Three surrogates (Tweedie single-step, marginal substitution, timestep-prior) make the bilevel objective tractable and **trainable in batch** where exact unrolling OOM'd on a single sample, raising attack success **~20%** over heuristic baselines at a **5% poison rate**.

Project: Compositional Targeted Multi-Label Universal Perturbations (CVPR 2025)

- **Problem**: Targeted universal perturbations for multi-label models require solving for every target label combination, which is **exponential in label count** and makes full-label-space evaluation infeasible on datasets such as OpenImages.

- *Solution:* Built a compositional optimization framework that decomposes targeted perturbations under a **label-independence assumption**, reducing 2^n combinations to n per-label problems: **141h to 48h (2.9×)** wall-clock on 100K OpenImages samples over a 100-label subset (on V100 32 GB). Perturbations for unseen label combinations are generated in **constant time at inference**.

Project: Semantic-Aware Multi-Label Adversarial Attacks (CVPR 2024)

- *Problem:* Existing multi-label attacks fail to scale to large label spaces and corrupt predictions on correlated non-target labels.
- *Solution:* Identified **negative gradient correlation** between target and correlated labels as the root cause; built a **knowledge-graph** guided framework that generates semantic-aware perturbations and resolves the gradient correlation, achieving **~20%** higher attack success rate and reduced runtime over prior attacks on large datasets (PASCAL-VOC, NUS-WIDE, OpenImages).

Project: Text-based Error Generation using LLM Prompt Tuning

- *Problem:* Generating realistic, valid error descriptions for procedural task videos is hard to control – standard **Llama-2** prompting yields invalid or low-quality errors with no reliable filter.
- *Solution:* Built a **prompt-tuning pipeline** using adversarial discrete optimization with an automated **Llama-2 verifier** supplying an external validity signal in the feedback loop, raising output validity by **~35%** over the prompting baseline.

Project: Model Responsibility Abdication Auditing

- Built a **multi-agent audit** on Anthropic's **Pettri** probing **responsibility abdication** – autonomous LLM agents over-trusting auditor tool output under time pressure and deferring their own judgment – across **5 custom scenarios** with **self-defined judge dimensions** scoring multi-turn transcripts.

Graduate Research Assistant

CBL Lab, University of Houston

Aug. 2019 – July 2020

Houston, TX

Project: Explainable and Privacy-Aware Deep Learning

- Built cardiovascular disease prediction models on the **MESA** cohort, improving **F1 by 6%** over prior approaches, with **SHAP** and **LIME** attribution analyses to surface clinically interpretable risk factors.
- Designed an affective computing framework using temporal gait modeling over skeletal keypoints data, avoiding face and appearance capture for behavioral analysis; contributed directly to a **\$100K** research grant award.

AI Researcher

Deep Learning Lab, National Center of Artificial Intelligence

July 2018 – Aug. 2019

Islamabad, Pakistan

Project: Table Structure Recognition using Graph Neural Networks

- *Problem:* Table structure recognition in document images fails to generalize across the varied, irregular layouts found in real-world document scans.
- *Solution:* Built a **Graph Neural Network** formulation modeling cell-to-cell structural relationships, improving accuracy **37%** over baselines. Open-sourced a synthetic tabular-document generator. ([[Link](#)]).

ENGINEERING SKILLS

- **Programming Languages:** Python, C/C++, Java, SQL
- **ML Frameworks/Tools:** PyTorch, HuggingFace (Transformers, Diffusers), TensorFlow, OpenCV, NumPy
- **Training & Systems:** distributed data-parallel training (DDP), memory profiling and optimization, Git, Docker, Linux, GCP, AWS

SERVICE

- Served as a reviewer for TPAMI journal, CVPR, AAAI, ICCV, and ICLR conferences.
- Supervised an undergraduate research intern for six months at University of Houston.

AWARDS

- Selected as one of the 190 students from a pool of 2500+ candidates across Pakistan for fully funded UGRAD semester exchange program sponsored by the U.S. Department of State (worth **≈ \$22,500**).
- Prime Minister National ICT R&D Merit Scholarship for fully funded undergraduate degree and monthly stipend for four years (worth **≈ \$12,000**).
- Bronze medal and merit scholarship for 3rd standing (in a total of **≈ 40,000** students) in computer science higher education exams in Pakistan - FBISE, Islamabad.